**Tribhuvan University**

**Faculty of Humanities and Social Sciences**

# E-LEARNING PLATFORM

**A PROJECT REPORT**

**Submitted to**

**Department of Computer Application**

**Butwal Kalika Campus**

***In partial fulfilment of the***

***requirements for the Bachelor in Computer Application***

Submitted by

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Date:2082/08/05

Under the Supervision of

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**Tribhuvan University**

**Faculty of Humanities and Social Sciences**

**Butwal Kalika Campus**

## Supervisor's Recommendation

I hereby recommend that this project prepared under my supervision by Bibek Dangal and Anupam Baral entitled **"E-Learning Platform"** in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

………………..

**SIGNATURE OF SUPERVISOR**

Nirmal Aryal

Humanities and Social Sciences

Butwal, Rupandehi

Butwal Kalika Campus

# ABSTRACT

The E-Learning Platform is a web-based system designed to enhance access to education and facilitate seamless learning experiences. Users can browse courses, access study materials, submit assignments, and track their progress directly from their devices. The frontend of the system is built using React, providing a fast, smooth, and user-friendly interface. The backend is developed using PHP, with data stored and managed using SQL. This platform eliminates the need for physical attendance, making education more accessible and convenient. It supports features such as secure login, course browsing, assignment submission, progress tracking, and real-time feedback. An admin panel allows educators to manage courses, monitor student performance, and update content efficiently.

**Keywords:** E-learning platform, React frontend, PHP backend, SQL database, course browsing, assignment submission, user authentication, admin panel, progress tracking, interactive learning.

# Acknowledgment

We would like to sincerely thank everyone who contributed to the successful completion of our project, the E-Learning System.

First and foremost, we express our heartfelt gratitude to our supervisor, Mr. Nirmal Aryal, for his invaluable guidance, constructive feedback, and continuous encouragement throughout the project. His expertise and insights were instrumental in shaping and completing this work successfully.

We are also deeply grateful to Butwal Kalika Campus and the Department of Computer Applications for providing us with the resources, opportunities, and a supportive environment that enabled us to turn our ideas into reality.

Finally, we recognize and appreciate the dedication and collaboration of our project team. The teamwork, perseverance, and shared efforts were crucial in bringing this project to completion.

Thank you all for your contributions and for being a part of this journey with us.

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# List of Abbreviations

| | |
|---|---|
| **DFD** | Data Flow Diagram |
| **ER** | Entity Relationship |
| **PHP** | Hypertext preprocessor |
| **MYSQL** | My Structured Query Language |
| **JS** | JavaScript |

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# Chapter 1: Introduction

## 1.1. INTRODUCTION

E-Learning delivers education through the Internet, networks, or standalone systems, enabling the online transfer of skills and knowledge. It primarily uses electronic media, especially the Internet, to facilitate interaction between students and teachers and to distribute course materials.

Market growth is driven by the need to reduce education costs, government support for online learning, and increased smartphone and Internet access. However, challenges include an abundance of free content and lack of awareness.

Basic IT skills are essential for effective E-Learning, which has evolved alongside humanity's shift into the Information and Knowledge Ages. E-Learning bridges the gap between work and learning, allowing employees to develop skills using familiar tools. This is especially valuable in fast-changing fields like healthcare, where ongoing training is crucial.

E-Learning is a digital platform offering flexible, real-time online courses in areas like Machine Learning and Programming. It supports learning anytime and anywhere via connected devices, providing both individual and group sessions for interactive education.

## 1.2. Problem Statement

The current situation is very limited to a few resources; students are unable to gain knowledge more than the lecture provides them. This in the end limits student's performance, because everything a student gets is collected from lectures in class. Here are some of the problems of current system:

Key problems include:

i. Students submit assignment to lecturers through hard copies or personal emails.
ii. Students only get help from lectures if the lecturers are in their office.
iii. Students are required to physically available in the classroom in order to gain knowledge thereby sacrificing all other responsibilities.
iv. Students are unable to share resources effectively and hold group discussions that are monitored or supervised by lectures.
v. New lecturers added to a course must upload or provide their own teaching materials.

## 1.3. Objectives

The primary objective of this project is to design, develop, and implement a centralized web-based E-Learning platform to overcome the limitations of the traditional learning system by providing a unified and interactive environment for students and lecturers.

The major objectives are:

i. To develop a user-friendly online platform that allows students to browse courses, access learning materials, and complete assignments efficiently.
ii. To enable teachers to manage courses, upload content, and track student progress accurately.
iii. To provide a seamless enrollment and course management system suitable for both student and lecturer.
iv. Creating a centralized resource repository for course materials

## 1.4. Scope

The E-Learning Platform is created to make learning easier and more accessible for students, teachers, and educational institutions. Through this platform, students can access their study materials anytime, download notes, take quizzes, submit assignments, and track their overall progress. They can also interact with teachers and classmates and receive important updates directly from the system.

For teachers, the platform provides a simple way to create quizzes, check assignments, and monitor how well students are performing. Administrators can manage user accounts, organize courses, and ensure that everything runs smoothly and securely.

Overall, the system aims to reduce the limitations of traditional classroom learning by offering a convenient, flexible, and user-friendly digital space that helps students learn better and helps teachers teach more efficiently.

## 1.5. Limitation

The E-Learning Platform depends on stable internet connectivity, which can be a challenge for users in areas with poor networks. It also requires basic digital skills, so students or teachers unfamiliar with technology may face difficulties. The system cannot fully replace face-to-face classroom interaction, especially for practical subjects. Additionally, advanced features like AI-based learning, plagiarism detection, and secure online exam monitoring are not included in this version.

Furthermore, the platform may face performance issues if many users access it at the same time without proper server scaling. Some multimedia content, such as high-quality videos, may load slowly on low-end devices. The system also relies on users regularly updating their materials and activities, which may cause delays if instructors are not consistent. In addition, the lack of integrated virtual classroom tools limits real-time interaction. These factors can affect the overall effectiveness of the learning experience.

## 1.6. Report Organization

The report is organized into five main chapters to provide a systematic and comprehensive overview of the E-Learning Platform. Each chapter focuses on specific aspects of the project, from background study to conclusion and recommendations.

**Chapter 1:** The introduction provides a general overview of the project, including the problem statement, objectives, scope, limitations, and report organization. This chapter highlights the purpose and importance of developing the system to improve access to education and enhance learning experiences for students and teachers.

**Chapter 2:** Background Study and Literature Review provide a theoretical foundation and describes fundamental concepts, terminologies, and technologies related to the project. It also reviews similar projects and relevant research conducted by other authors to identify best practices and existing gaps.

**Chapter 3:** System Analysis and Design discusses the detailed analysis and design of the system. It includes requirement analysis (functional and non-functional), feasibility study (technical, operational, economic, and schedule), data modeling using ER diagrams, process modeling using DFDs, system architecture, database schema design, and user interface design.

**Chapter 4:** Implementation and Testing describes the tools, programming languages, and platforms used to develop the system. It also presents the implementation details of each module and the testing procedures, including unit testing and system testing, to ensure functionality and reliability.

**Chapter 5:** Conclusion and Future Recommendations summarizes the outcomes and lessons learned from the project. It highlights the system's contributions and provides suggestions for future enhancements and improvements.

# Chapter 2: Background Study and Literature Review

## 2.1. Background Study

The rapid advancement of information and communication technology (ICT) has greatly transformed many industries, including education. Traditional classroom-based learning methods often face challenges such as limited accessibility, fixed schedules, and resource constraints, which can restrict students' opportunities to learn effectively. Additionally, physical classrooms sometimes limit personalized learning and timely feedback.

In response to these challenges, E-Learning Platform have become a popular and effective solution. These platforms provide a digital environment where students can access course materials, watch video lectures, complete assignments, and participate in quizzes anytime and anywhere. E-Learning not only offers flexibility and convenience for learners but also helps educators manage courses efficiently, track student progress, and provide timely support.

The integration of modern web technologies, such as React for frontend development, Node.js or PHP for backend processing, and SQL databases for data management, has enabled the creation of scalable, interactive, and user-friendly e-learning platforms. For educational institutions, especially those with limited physical infrastructure, such systems offer a chance to broaden their reach, improve learning outcomes, and meet the growing demand for accessible and technology-driven education.

For educational institutions, especially those with limited physical infrastructure, such systems offer a chance to broaden their reach, improve learning outcomes, and meet the growing demand for accessible and technology-driven education. As digital learning continues to evolve, E-Learning platforms play a crucial role in supporting lifelong learning and adapting education to the needs of a rapidly changing technological world.

## 2.2. Literature Review

E-learning platforms have revolutionized higher education by creating dynamic and accessible environments that boost student engagement and academic performance. Research shows that platforms like Moodle and iSpring provide interactive learning spaces, with 78% of students reporting no difficulties using them, indicating high usability. These platforms also offer cost-effective education delivery by centralizing resources, aligning well with E-learning's goal of unified content access. However, as noted by 17% of students, continuous updates are necessary to keep pace with evolving technology. Systematic reviews emphasize that when implemented correctly, e-learning platforms enable effective virtual interactions that foster knowledge construction and meet educational demands in dynamic contexts, such as those prompted by the COVID-19 pandemic. E-learning builds on these insights by offering a scalable platform that supports adaptive learning and seamless access to digital resources.

Effective teacher-student interaction and the development of digital competencies are vital to e-learning success. Studies reveal that virtual platforms improve communication, with 40.1% of students finding it easy to reach instructors and 42.3% rating virtual teaching positively. Collaborative approaches and strong tutor-learner relationships encourage meaningful learning and knowledge creation. These platforms facilitate continuous communication—key to E-learning's real-time discussion forums and assignment feedback—helping to overcome limitations of traditional in-person settings. Additionally, e-learning fosters digital skills development, with blended learning proving effective for practical subjects and gamification boosting motivation. E-learning incorporates these findings by providing an intuitive interface that supports ICT skill growth and interactive course creation while addressing challenges like connectivity and platform familiarity.

# Chapter 3: System Analysis and Design

## 3.1. System Analysis

System analysis involves studying the requirements and structure of the E-Learning Platform to ensure it meets the needs of both students and instructors. This process includes identifying functional and non-functional requirements, assessing system feasibility, and modeling data and processes to guide the system's design and implementation effectively.

### 3.1.1. Requirement Analysis

The requirement analysis defines the features and constraints of the system. It is divided into functional requirements and non-functional requirements**.**

**I. Functional Requirements**

Functional requirements describe the specific behaviors and functionalities that the system must provide to its users. The main functional requirements for the E-Learning Platform are:

i. CRUD operations for users, courses, materials, assignments, and quizzes.
ii. Students can enroll in courses, access study resources, submit assignments, and quizzes.
iii. Lecturer can create and manage courses, upload resources, post assignments, and conduct quizzes.
iv. Centralized repository for resources.
v. Secure login with role-based access for students, lecturer, and administrators.

**ii. Non-Functional Requirements**

i. User-friendly, intuitive interface.
ii. High availability and accessibility.
iii. Efficient performance for fast loading.
iv. Secure data protection.

# Use Case Diagram

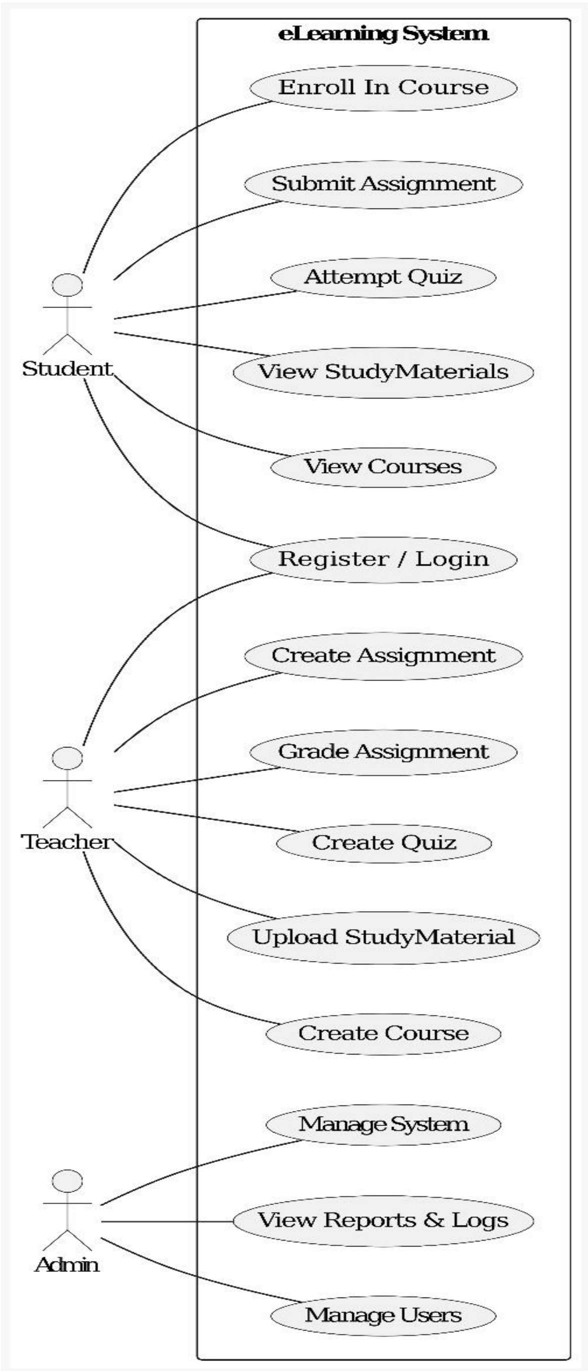


**Figure 3.1: Use Case Diagram**

### 3.1.2. Feasibility Analysis

Feasibility analysis evaluates whether the E-learning Platform can be successfully developed and implemented, considering technical, operational, economic, and scheduling aspects.

i. **Technical Feasibility**

   The system uses widely accepted and reliable technologies, including React for the frontend, PHP (or Node.js) for backend processing, and SQL for database management. These technologies are compatible with standard web servers and offer scalability and performance suitable for educational institutions of various sizes. The necessary hardware and software resources are readily available, and the development team has the required technical expertise. Therefore, the project is technically feasible.

ii. **Operational Feasibility**

   The system is designed to be user-friendly, allowing both students and instructors to use it with minimal training. Students can easily access course materials, submit assignments, and participate in quizzes, while instructors can manage courses, monitor progress, and provide feedback efficiently. The platform addresses challenges of traditional learning methods by improving accessibility and communication, making it operationally feasible.

iii. **Economic Feasibility**

   The project is cost-effective since it relies mainly on open-source technologies, minimizing software licensing fees. It requires minimal additional hardware investment and helps reduce administrative overhead while enhancing learning outcomes. The benefits of improved accessibility, streamlined course management, and better student engagement outweigh the development and maintenance costs, making the system economically feasible.

iv. **Schedule Feasibility**

   Given the project's defined scope and the use of modern development tools, the E-Learning platform can be developed and deployed within a reasonable timeframe. A clear project plan and focused development phases will ensure timely completion and delivery. Therefore, the project is schedule feasible.

### 3.1.3. Data Modeling (ER-Diagram)

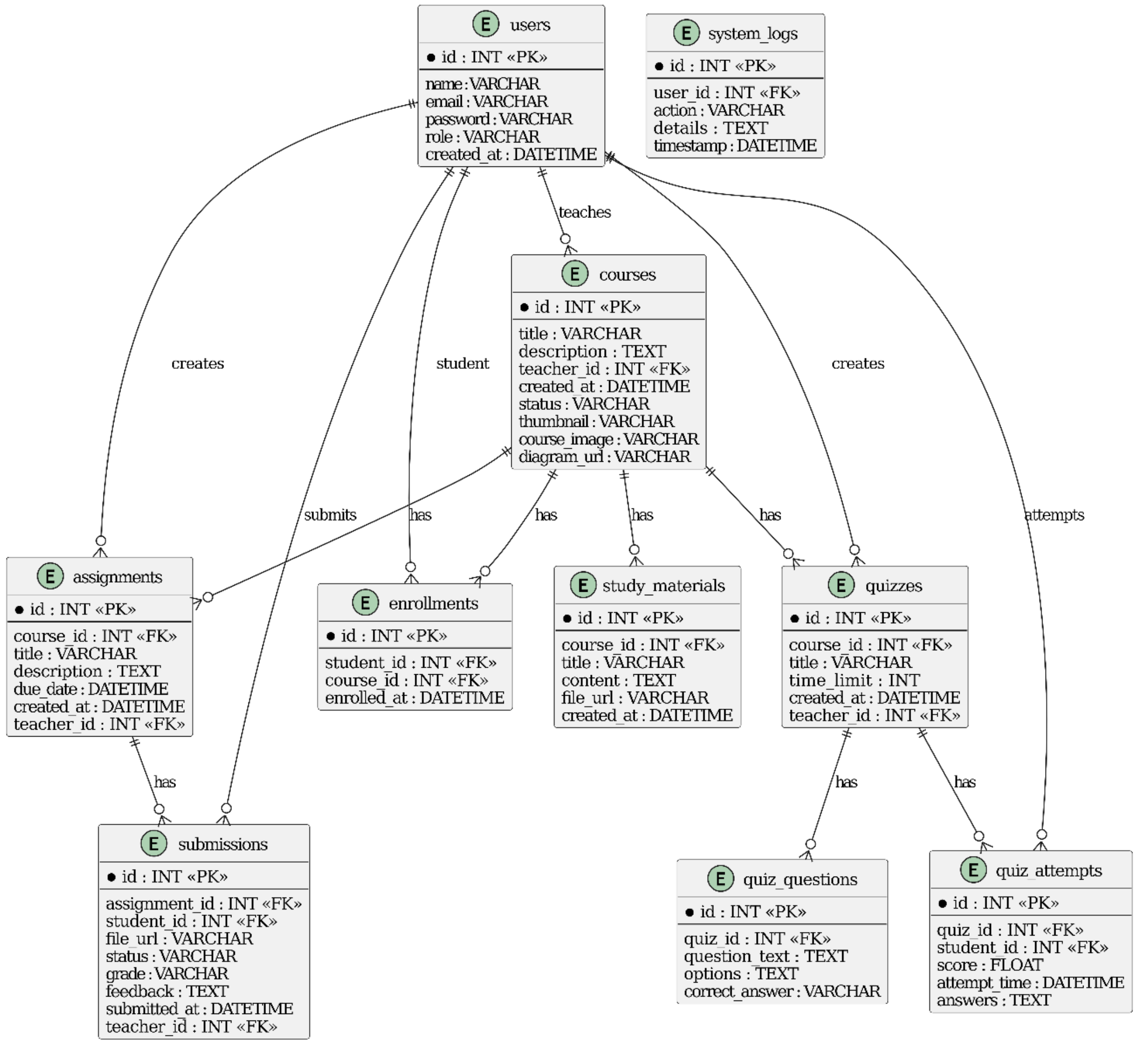


**Figure 3.2: ER-Diagram**

### 3.1.4. Process Modeling (DFD)

Process modeling illustrates how data flows through the E-learning platform. A Data Flow Diagram (DFD) is used to represent the processes, data stores, external entities, and the movement of information within the system. The DFD helps visualize how inputs are transformed into outputs and clarifies system functionality at different abstraction levels.

i. **Level 0 DFD (Context Diagram)**

The Level 0 DFD represents the system as a single process with external entities interacting with it. It provides a high-level overview of data flow between the system and its users.

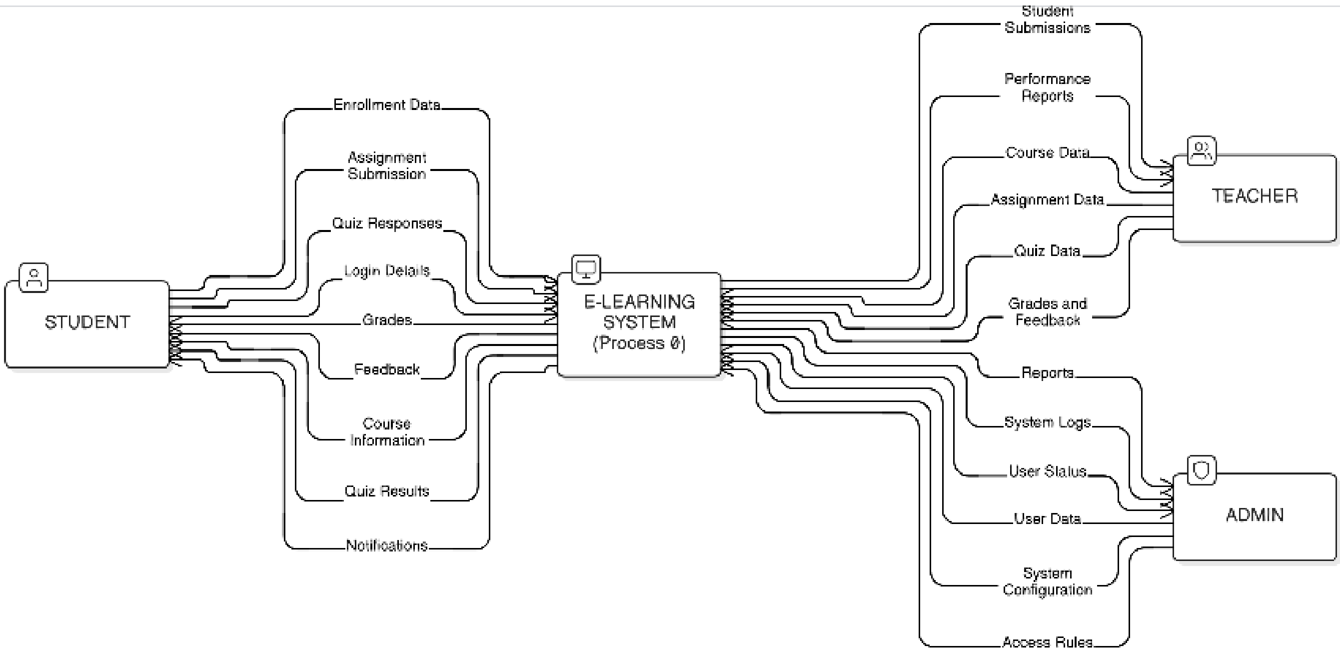


**Figure 3.3: Level 0 DFD**

### ii. Level 1 DFD

The Level 1 DFD breaks the main system into key processes and shows how data moves between them.

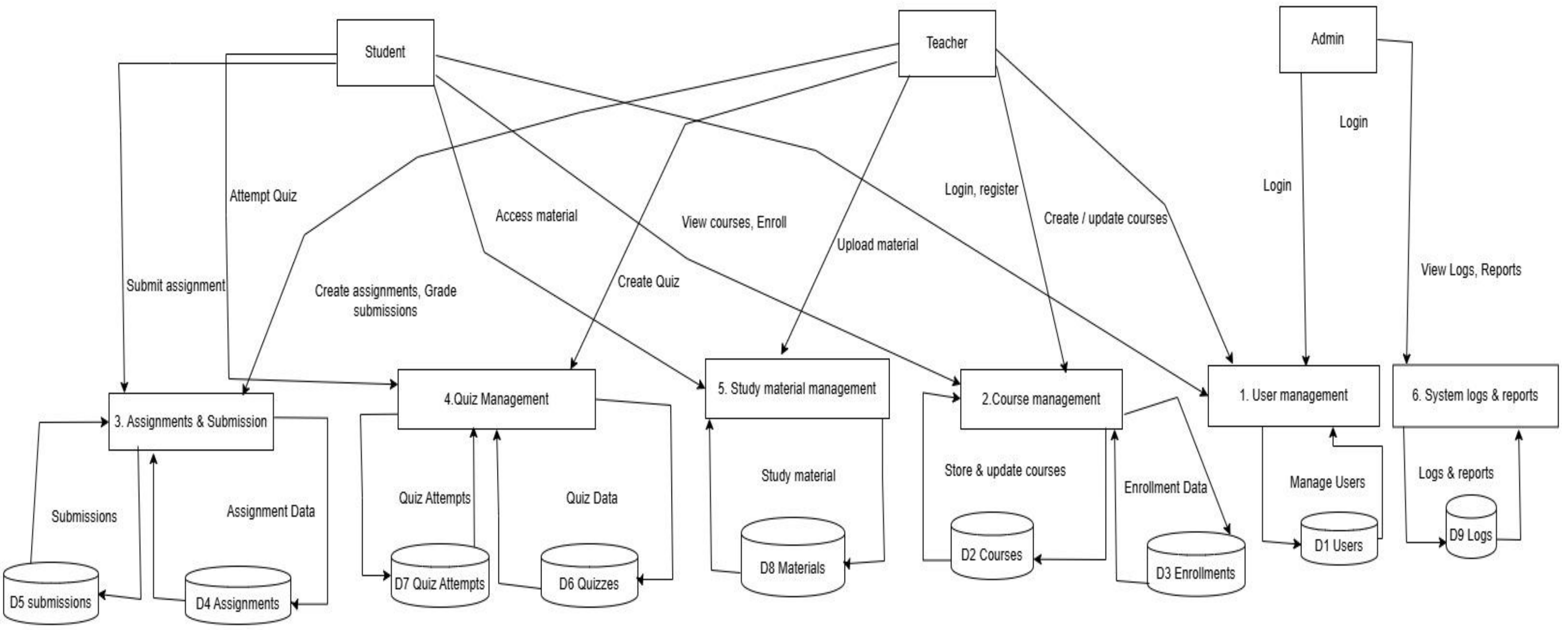


**Figure 3.4: Level 1 DFD**

## 3.2. System Design

System design defines how the E-Learning Platform is structured, how its components interact, and how data is managed. It focuses on converting the system requirements into a technical blueprint for implementation. This stage outlines the overall system architecture, including the frontend, backend, and database layers, ensuring that each component functions cohesively.

### 3.2.1 Architectural Design

The architecture of the E-Learning System follows a three-tier architecture, which separates the application into:

- **The Presentation Layer**, developed using React.js, manages the entire user interface for both students and instructors. It includes pages for course browsing, video lectures, assignments, quizzes, progress tracking, and instructor dashboards. This layer communicates with the backend exclusively through API requests, ensuring smooth interaction and a responsive user experience.
- **The Application Layer**, built using PHP (or Node.js), handles server-side logic such as user authentication, course and content management, assignment submission, grading, and feedback. It validates incoming data, enforces business rules, and provides structured API responses for the frontend.
- **The Database Layer**, implemented in MySQL, stores all persistent system data, including user profiles, course details, learning materials, assignments, quiz results, and activity logs. It ensures data integrity and maintains relationships among entities.

Together, these three layers create a modular, scalable, and efficient architecture for the E-Learning System. This layered approach ensures a clear separation of concerns, allowing each layer to be developed, tested, and maintained independently. It enhances system reliability by isolating changes in one layer from affecting others, thereby reducing complexity during updates or upgrades.

### 3.2.2. Database Schema Design

The database for the E-learning Platform, named **elearning_db**, is designed to store and manage all relevant data efficiently. It uses MySQL and follows a relational model to ensure data consistency, integrity, and proper relationships among different entities.

The **elearning_db** database contains tables for key e-learning components like users, courses, lessons, enrollments, and assessments, ensuring efficient data management and scalability.

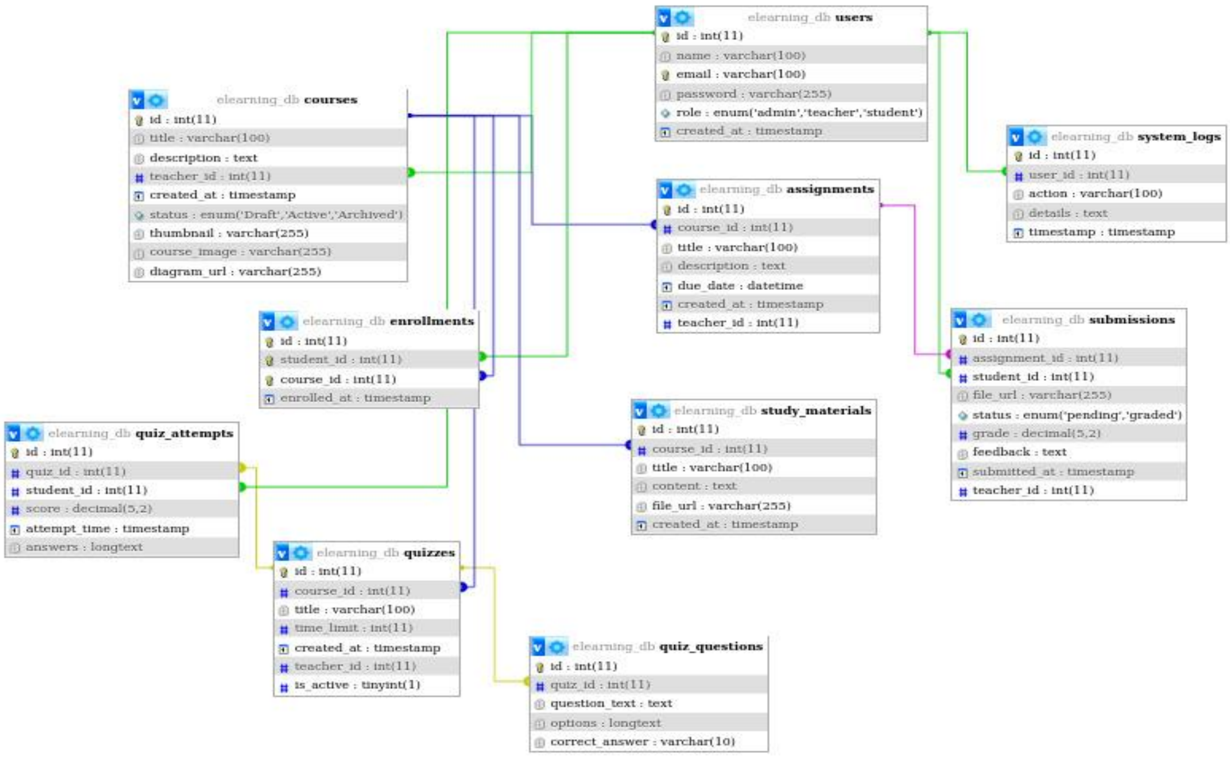


**Figure 3.5: Database Schema**

### 3.2.3. Interface Design

i. The interface design of the E-Learning Platform focuses on providing a clear, intuitive, and user-friendly experience for both students and instructors. The design ensures that users can navigate easily, access required learning materials, and perform all tasks efficiently.

ii. **Student Interface**

The student interface is designed to be visually appealing, organized, and responsive. The Home Page displays available courses and announcements. The Course Page lists all course contents, notes, assignments, and quizzes. Students can open each module to watch lessons or download study materials. The Assignment Page allows learners to upload submissions, while the Quiz Page provides an easy-to-use interface for taking online quizzes. The Progress Tracking Page shows completed tasks, grades, and overall course progress, helping students monitor their learning journey.

iii. **Instructor Interface**

The instructor interface provides full control over course management. The Dashboard displays a summary of total students, submitted assignments, upcoming deadlines, and course activity. The Course Management Page allows instructors to create, update, or delete course modules, upload note, and organize content. The Assignment Management Page enables teachers to create assignments, review submissions, and give grades or feedback. The Quiz Management Page allows setting quiz questions, configuring timing, and reviewing quiz results. Additionally, a Student Performance Page provides insights into student progress and participation.

iv. **Interface Structure**

Both student and instructor interfaces follow a consistent layout with clear navigation menus, buttons and well-structured content sections to ensure smooth interactions. The system is fully responsive, making it accessible on desktops, tablets, and mobile devices. Validation messages, alerts and clear labels are included to reduce errors, improve usability, and enhance overall user satisfaction.

### 3.2.4. Physical DFD

The Physical DFD represents how the proposed E-learning Platform operates in the real environment, showing actual hardware, software, people, and physical media involved in data processing. It illustrates who performs the processes, where the processes occur, and how the data flows through physical components of the system.

In this physical DFD:

i. **External entities** such as instructors and students, interact with the system using devices like smartphones, laptops, or desktop computers.
ii. **Processes** represent real-world actions performed by the system modules, such as User Authentication, Course Management, Assignment Handling, and Quiz Processing.
iii. **Data stores** represent physical storage such as the Database Server for storing instructor details, student profiles, course materials, assignments, quiz results, and progress records.
iv. **Data flows** show the movement of information between external entities and the system over the internet.

# Chapter 4: Implementation and Testing

## 4.1. Implementation

### 4.1.1. Tools Used

The development of the E-learning Platform required a combination of programming languages, frameworks, and database platforms. The following tools and technologies were used:

**Table 4.1: Tools Used**

| Tools Name: | Purpose: |
|---|---|
| VS Code | Used to develop and manage frontend and backend source code. |
| XAMPP Server | Provides local server environment for running PHP and MySQL |
| MySQL | Stores and manages all project data such as users, courses, study materials, quizzes, and progress records. |
| Git & GitHub | Version control and code hosting for project backup and deployment. |

1. **CASE Tools**
   i. Draw.io: Used to design ER diagrams, use case diagrams, DFDs, and architectural diagrams.
   ii. Git & GitHub: Used for version control, collaboration, and maintaining development history.
2. **Programming Languages**
   i. It is used to build interactive and dynamic user interfaces, providing reusable components for pages such as the course list, course details, assignment submissions, quizzes, and the admin dashboard.
   ii. Javascript: Used throughout the frontend for API calls, state management, and handling UI actions.
   iii. PHP (Backend): Handles server-side logic, request processing, form validation, and communication with the database.
3. **Database Platforms**
   i. MySQL: Used as the primary relational database to store users, courses, lessons, assignments, quiz results, and instructor details.

### 4.1.2. Implementation Details of Modules

The entire project mainly consists of 2 modules, which are:

1. **Admin Module**
   i. Login and signup
   ii. Add, delete, view, and update courses and study materials
   iii. View student details and enrollment status
   iv. Manage assignments, quizzes, and grading
2. **Student Module**
   i. Signup and login
   ii. Browse and view available courses
   iii. Enroll in courses
   iv. Access lessons and study materials
   v. Submit assignments and attempt quizzes
   vi. Track course progress and view grades

## 4.2. Testing

Testing is an essential phase of the development process to ensure the E-Learning Platform functions correctly, meets requirements, and provides a smooth user experience. Both **Unit Testing** and **System Testing** were performed to validate individual components as well as the overall system behavior.

### 4.2.1. Test Cases for Unit Testing

**Table 4.2: Unit Test Cases for E-Learning Platform**

| S. N | Module | Test Description | Input | Expected Output |
|---|---|---|---|---|
| 1 | Login Function | Check valid login | Correct email & password | User logged in successfully |
| 2 | Login Function | Check invalid login | Wrong email/password | Error message displayed |
| 3 | Course Enrollment | Test enrolling in course | Select a course | Course added to student dashboard |
| 4 | Upload content (Instructor) | Add new lecture/notes | Title, file upload, description | Content uploaded successfully |

| | | | | |
|---|---|---|---|---|
| 5 | Assignment Submission | Submit assignment | File upload + comments | Content uploaded successfully |
| 6 | Quiz Module | Attempt quiz | Select answers + submit | Quiz auto-graded and results generated |
| 7 | Grade Update (Instructor) | Update student grades | Marks + feedback | Grades updated in database |

### 4.2.2. Test Cases for System Testing

**Table 4.3: System Test Case for E-Learning Platform**

| S. N | Scenario | Test Description | Expected Output |
|---|---|---|---|
| 1 | User Registration | User creates a new account | Account created successfully |
| 2 | Course Browsing | User views available courses | Courses load with proper categories |
| 3 | Enrolling in course | Student enrolls in selected course | Course added to student profile |
| 4 | Submitting Assignment | Student uploads assignment file | Status = Submitted and visible to instructor |
| 5 | Quiz Attempt | Student attempts quiz inside the course | Quiz submitted & score generated |
| 6 | Instructor Grades Assignment | Teacher uploads notes/PDFs/images | Materials appear in student course section |
| 7 | Creating / Editing Course Material | Instructor uploads or edits lectures, notes, or quizzes | Updates visible to enrolled students |
| 8 | Progress Tracking | Student checks progress of completed lessons | Updated progress percentage shown |

# Chapter 5: Conclusion and Future Recommendations

## 5.1. Lesson Learnt

During the development of the **E-Learning Platform**, several valuable lessons were learned. The project enhanced our understanding of how frontend and backend systems communicate using APIs and how databases store and manage user information, courses, and learning materials. We gained practical knowledge of implementing CRUD operations in a real application and strengthened our skills using tools such as **React, PHP, and MySQL**.

Additionally, designing ER Diagrams, Data Flow Diagrams, and improved our understanding of system analysis and design concepts. The project also helped enhance problem-solving abilities, teamwork, planning, version control usage, and proper testing methods. Overall, the development experience strengthened our technical and analytical skills for building real-world web applications.

## 5.2. Conclusion

The **E-Learning Platform** was successfully developed to simplify the teaching and learning process by providing a digital platform for students and teachers. The system met the functional requirements defined in the proposal and operated smoothly with minimal issues. Although there were challenges during the development and integration phases, continuous testing, feedback, and improvements enabled us to overcome them effectively.

Throughout the project, we gained deeper knowledge of **frontend–backend communication**, database structuring, API handling, user authentication, and responsive UI development. We also improved our ability to perform requirement analysis, create system models, and transform them into a fully working application using modern technologies.

The system provides essential e-learning features such as **course browsing, enrollment, lesson viewing, study material access, quizzes, and progress tracking**. Although the current system is designed for small-scale use, it can be expanded with additional functionalities like discussions, notifications, and real-time communication to support a larger user base in the future.

## 5.3. Future Recommendations

To make the E-Learning Platform more efficient, user-friendly, and engaging, the following improvements are suggested:

i. **Introduce AI-Based Recommendations:** Suggest courses, quizzes, or study materials based on a learner's interests, performance, and learning history to create a personalized learning experience.

ii. **Add Multi-Instructor Support:** Allow multiple teachers to upload courses, manage content, and interact with their students through a shared platform.

iii. **Provide Certificates & Badges:** Offer digital certificates and achievement badges to increase motivation and recognize student accomplishments.

iv. **Add Multi-Language Support:** Enable Nepali, Hindi, or other language options to make the platform accessible to a wider range of learners.

v. **Add Notification system**: Notify users about new courses, assignment deadlines, quiz results, and announcements through email, SMS, or in-app notifications.

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# Appendices

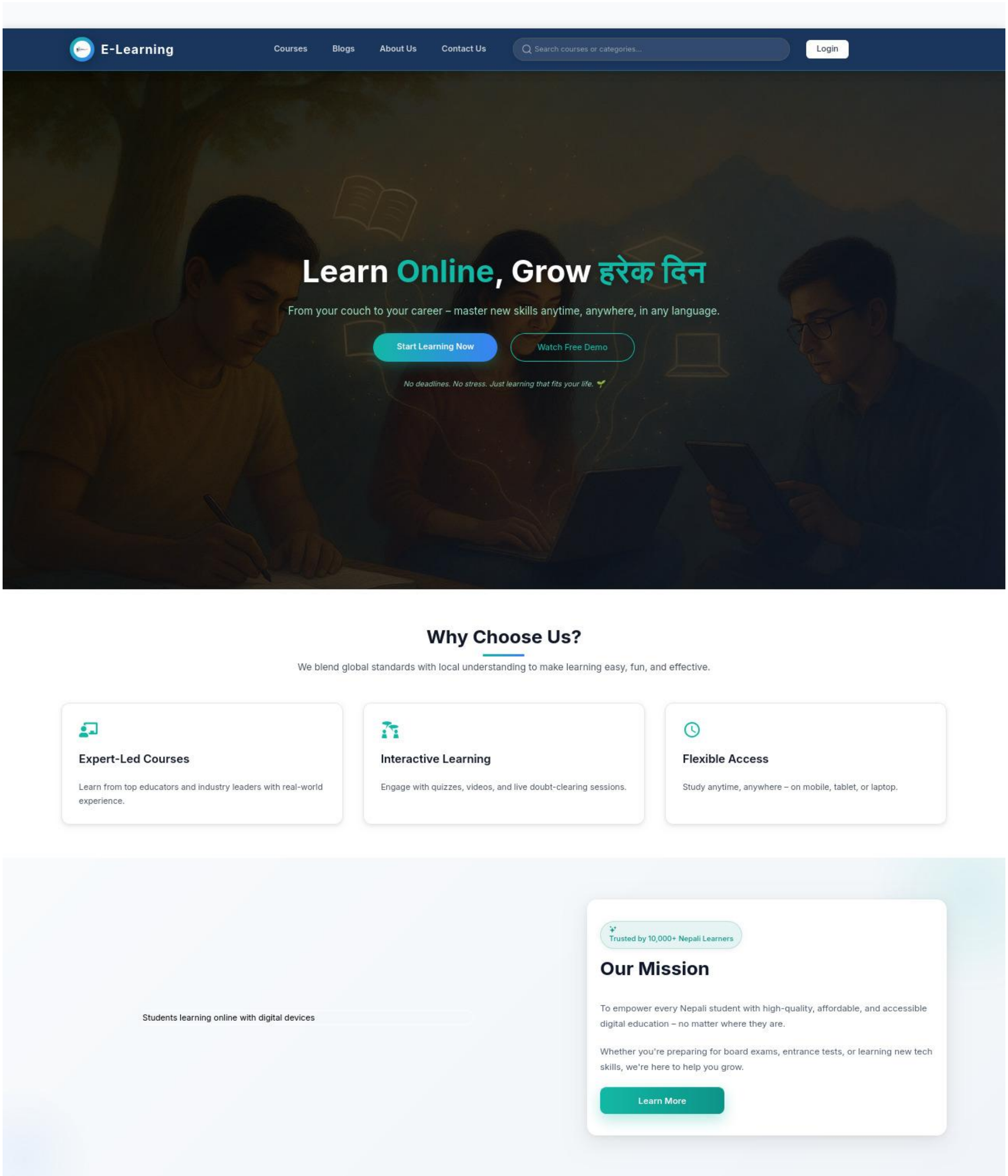


**Figure A1: Landing page of E-Learning Platform**

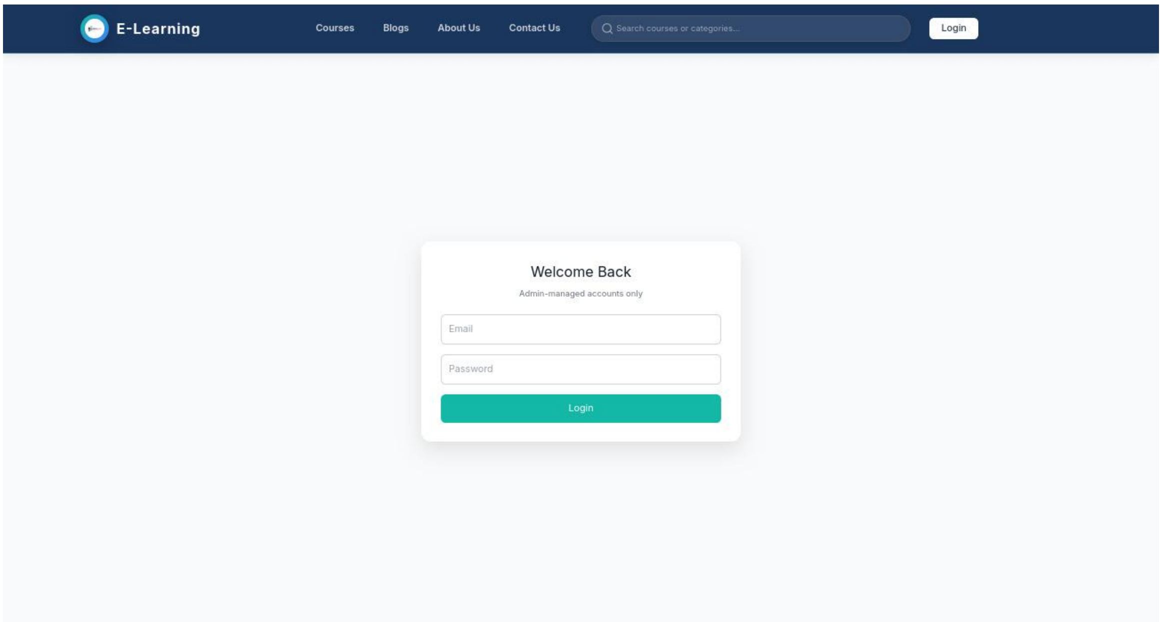


**Figure A2: Login Page**

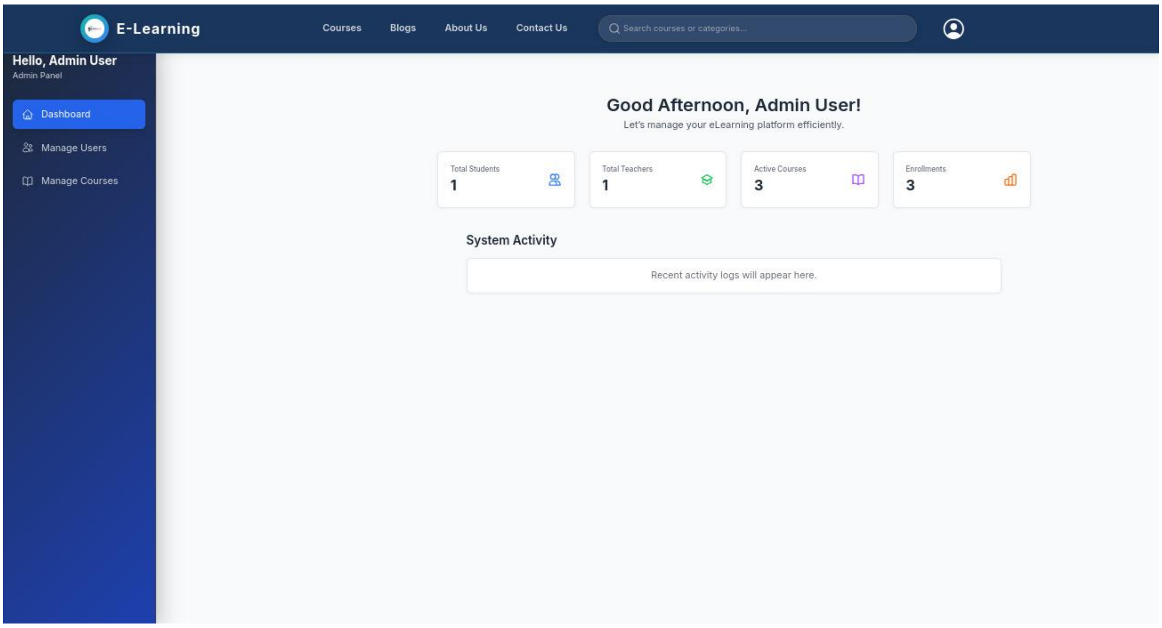


**Figure A3: Admin dashboard**

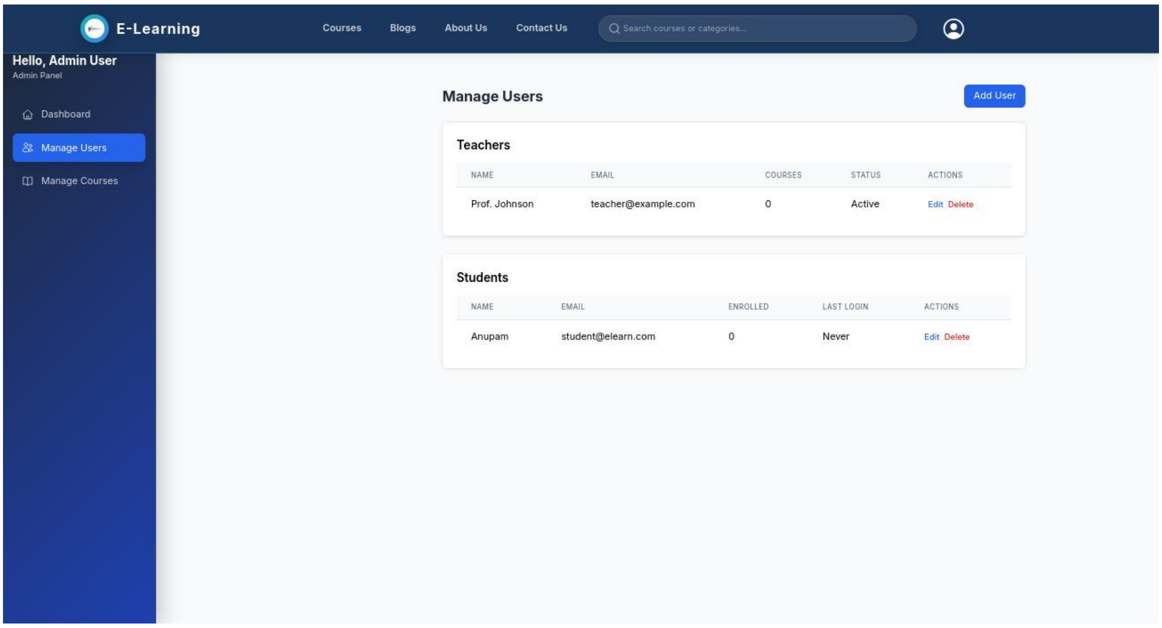


**Figure A4: Admin managing user**

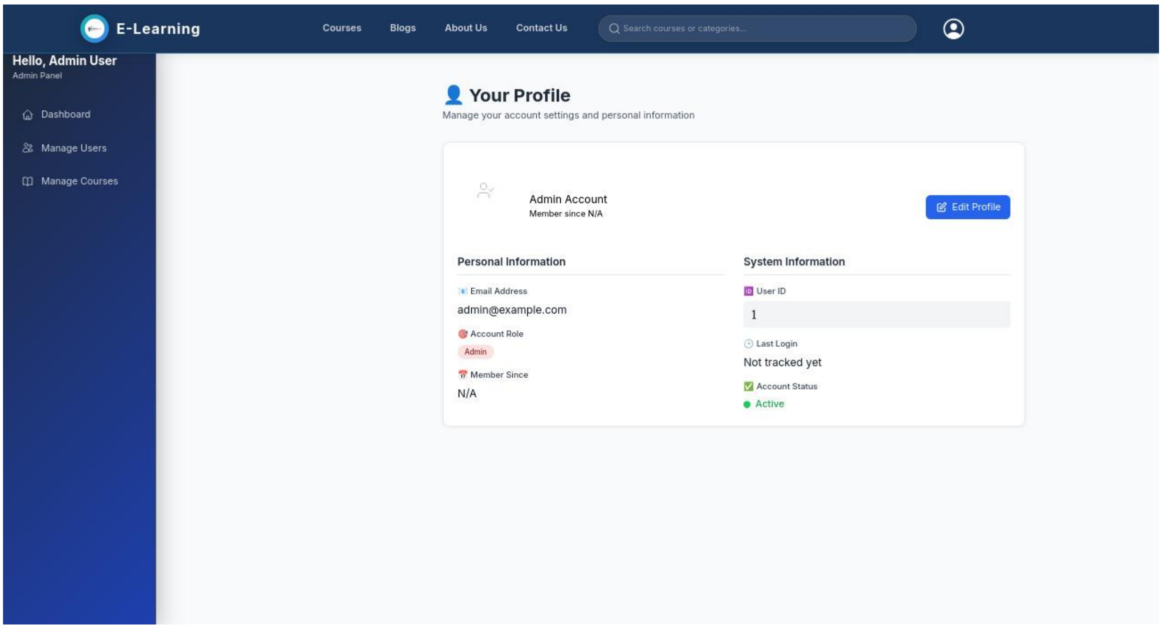


**Figure A5: Admin profile**

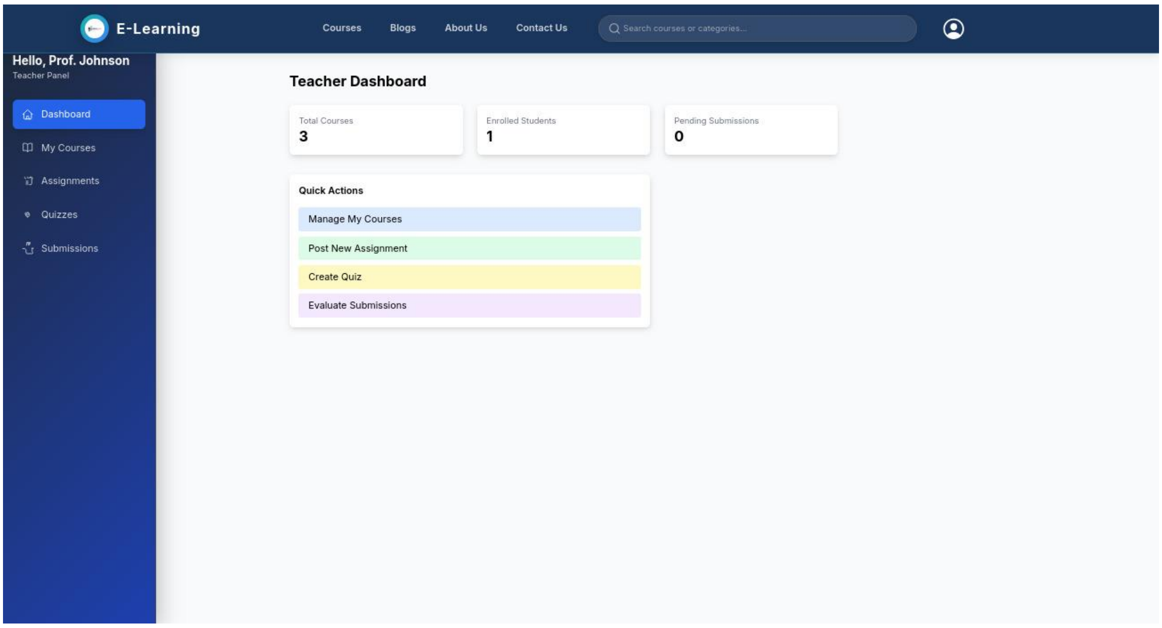


**Figure A6: Teacher Dashboard**

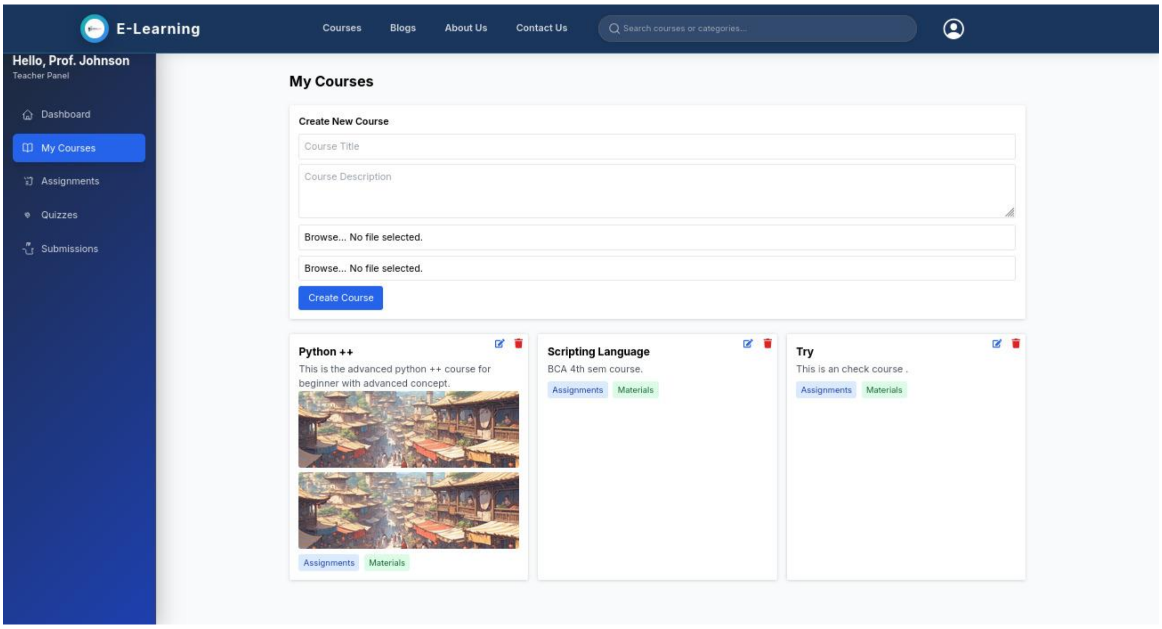


**Figure A7: Teacher Course Creation**

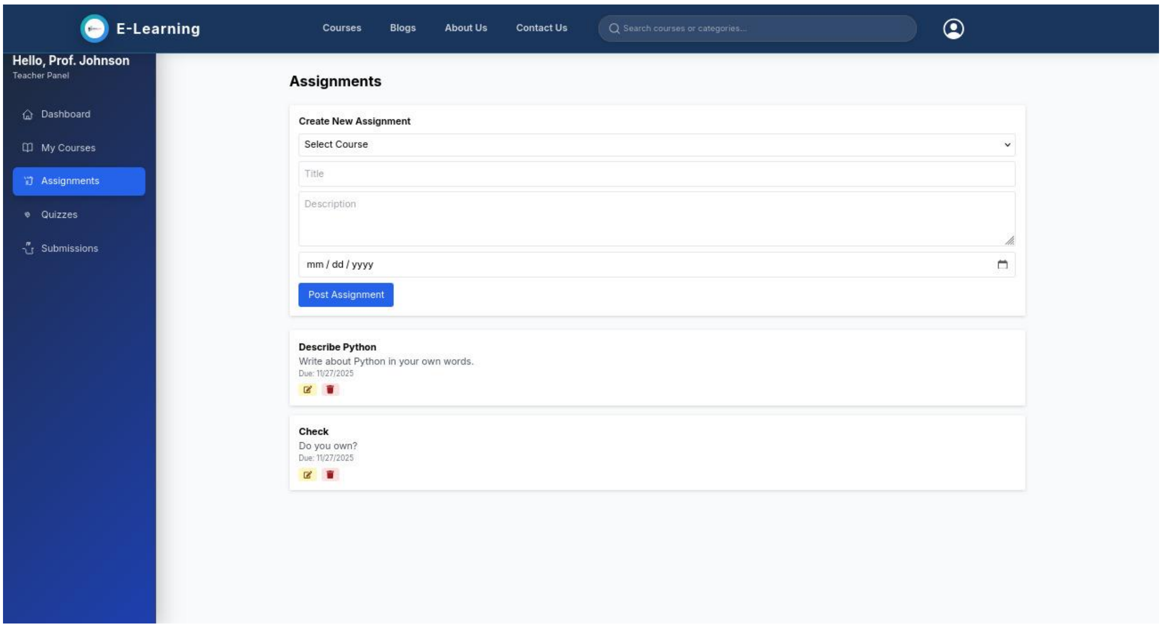


**Figure A8: Teacher Assignment Creation**

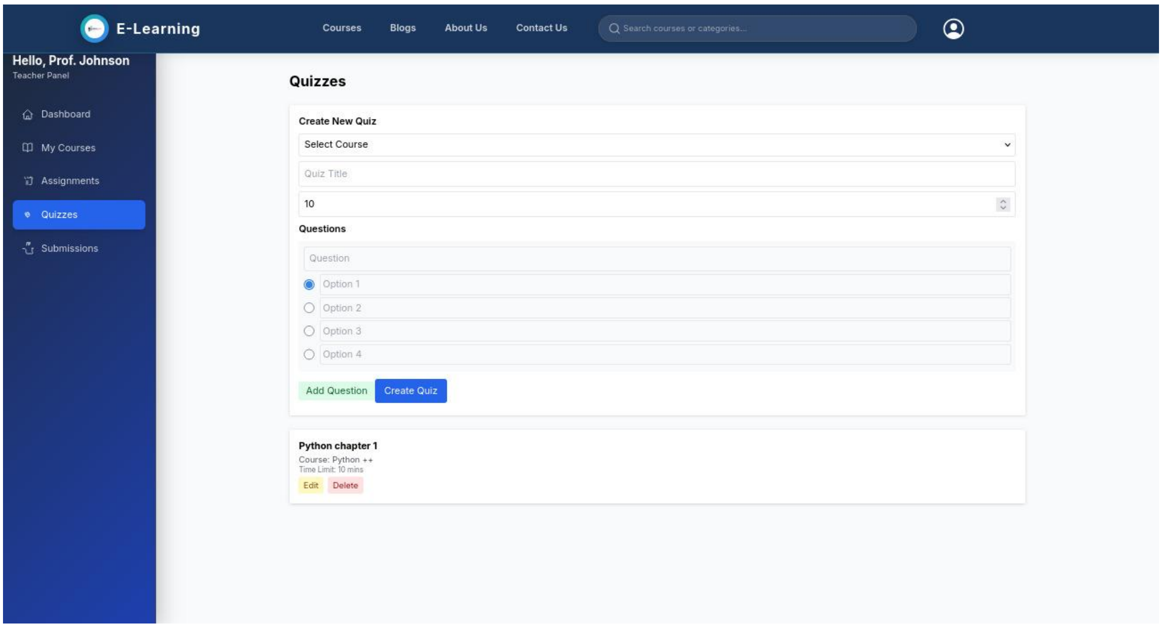


**Figure A9: Teacher Quiz Creation**

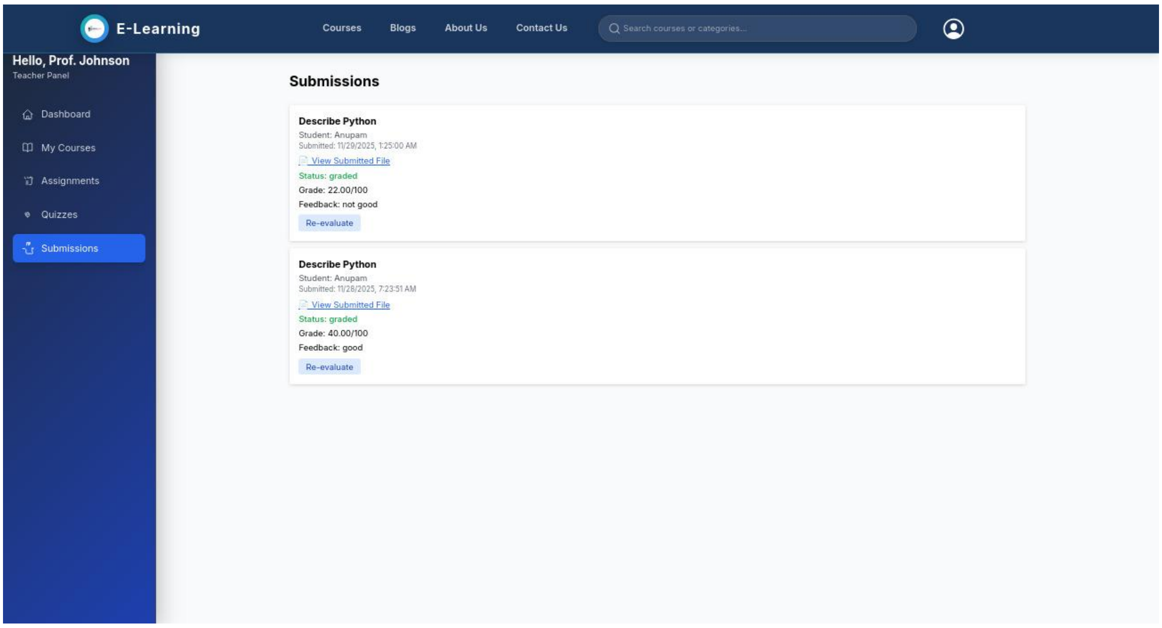


**Figure A10: Teacher Assignment Submission**

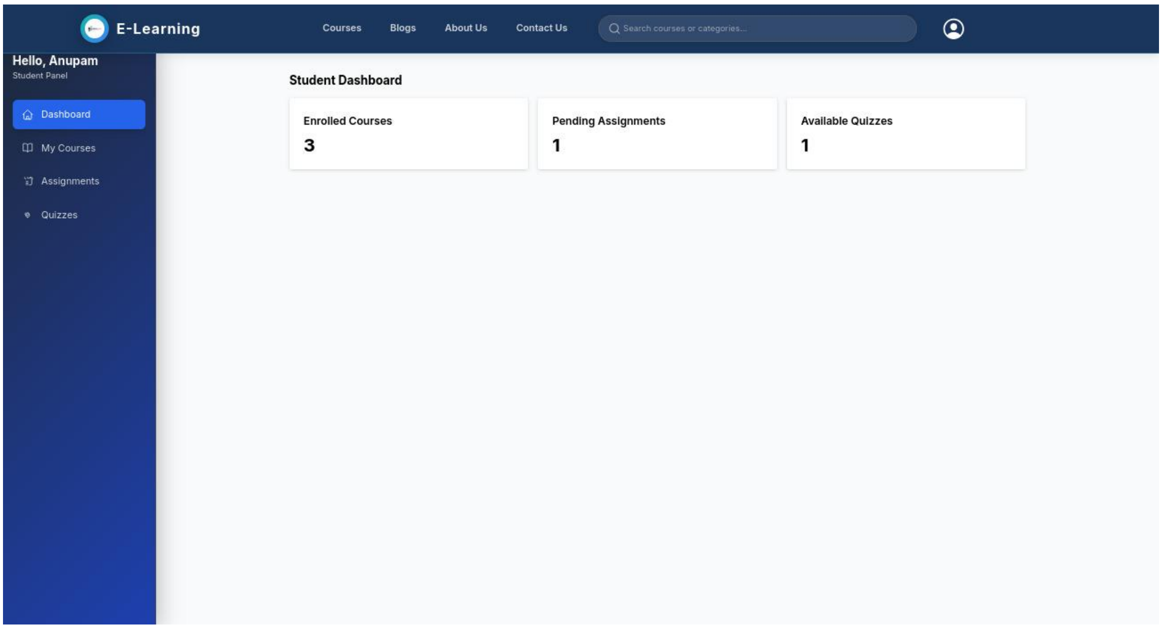


**Figure A11: Student dashboard**

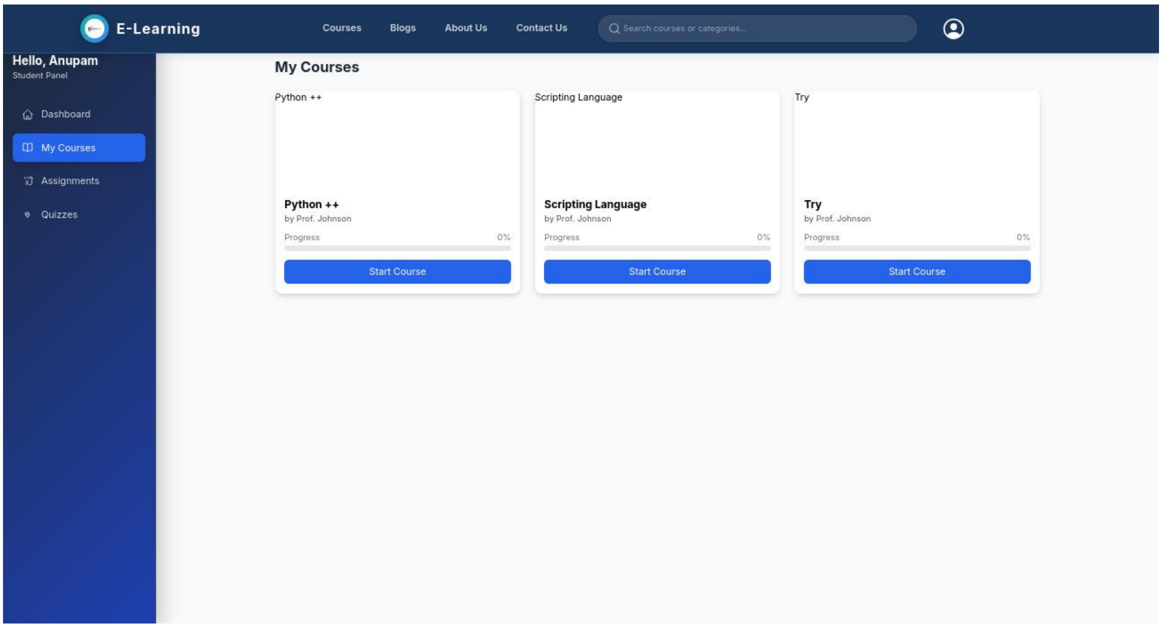


Figure A12: Student Enrolled Course

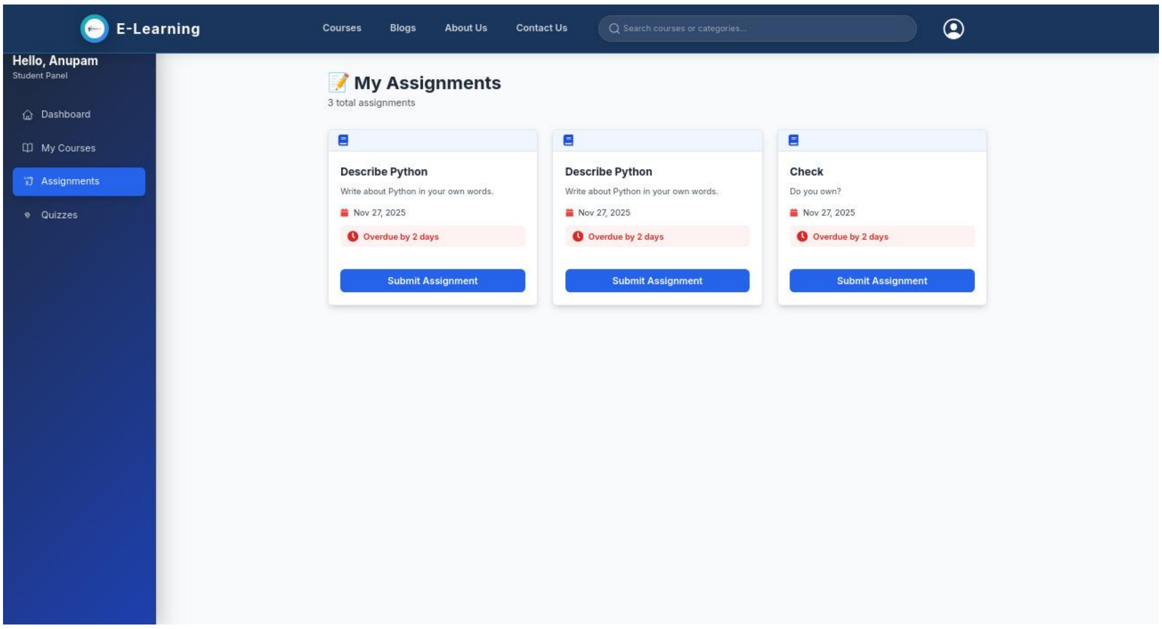


Figure A13: Student Assignment Submission

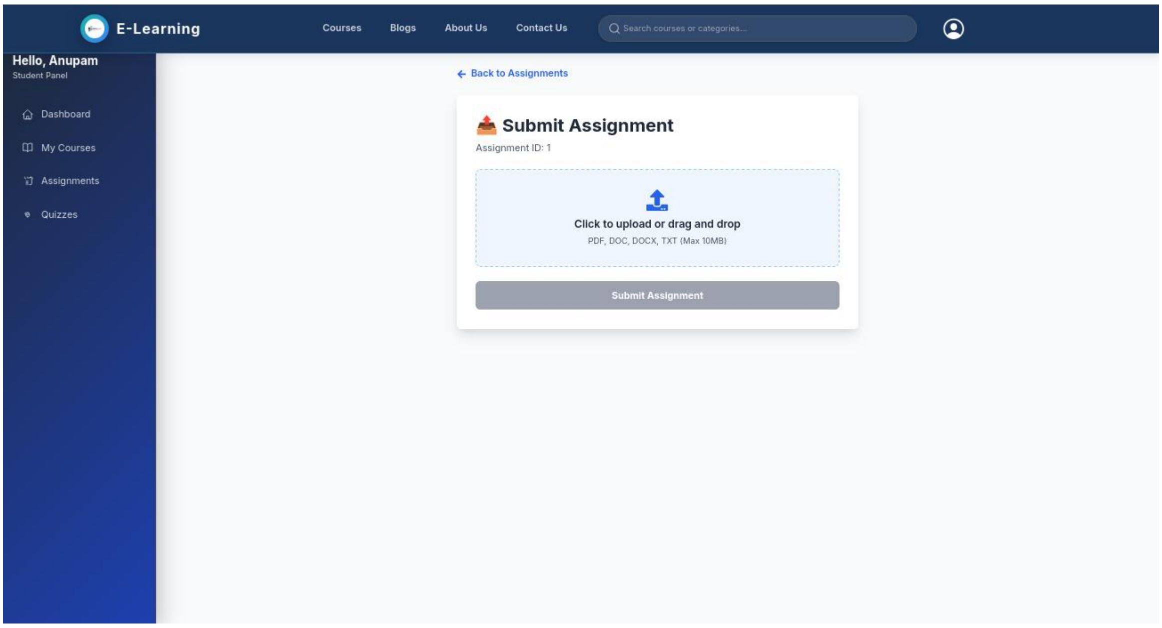


Figure A14: Assignment Submission